(No) effects of shadowing on predictive eye movements during non-dominant language comprehension

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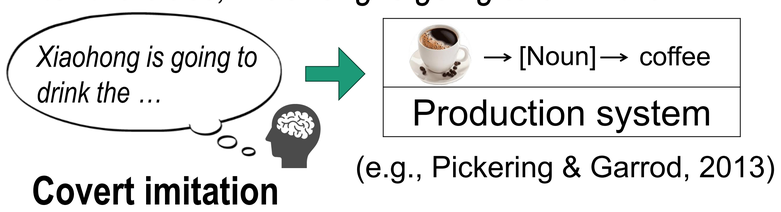
Background

- **Prediction-by-production**: Listeners use their production system for prediction (Huettig, 2015; Pickering & Gambi, 2018)
- This mechanism is optional and may be less engaged during non-dominant (vs. dominant) language processing

Prediction-by-production mechanism

After the class, Xiaohong is going to drink the …



(e.g., Pickering & Garrod, 2013)

Shadowing

- **RQ**: Does shadowing enhance prediction in a non-dominant language?

Experiment predictions

- If overt production enhances prediction, we expect a stronger predictability effect in the shadowing (vs. listening) task
- If it interferes with prediction (because overt production requires extra cognitive resources), we expect a weaker predictability effect in the shadowing (vs. listening task

Methods: Visual world paradigm

- 2 (Predictability) × 2 (Task) within-subjects design
- 80 English-Chinese early bilinguals, 28 items

Predictability

Predictable: 下课之后，小红要去**喝**那边的**咖啡**。
'After the class, Xiaohong is going to **drink** the **coffee** there.'

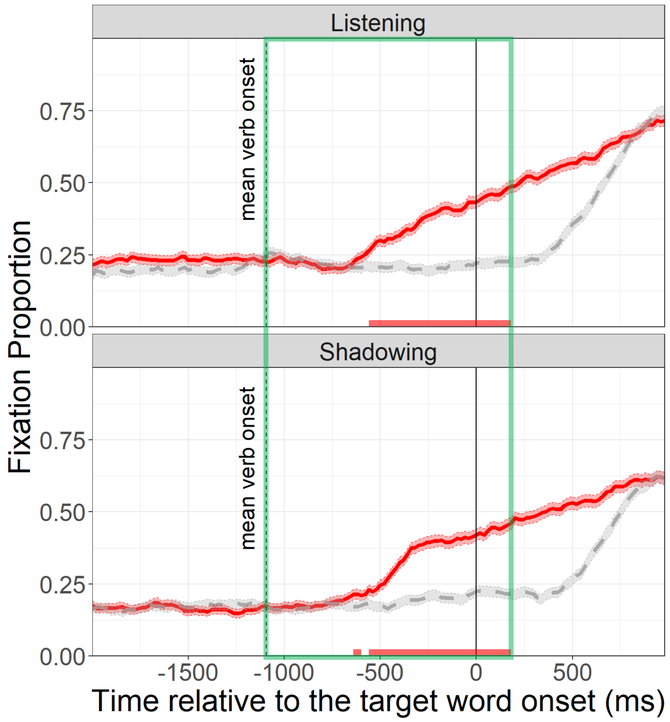
Unpredictable: 出门之前，小红要去**拿**那边的**咖啡**。
'Before going out, Xiaohong is going to **take** the **coffee** there.'

Task

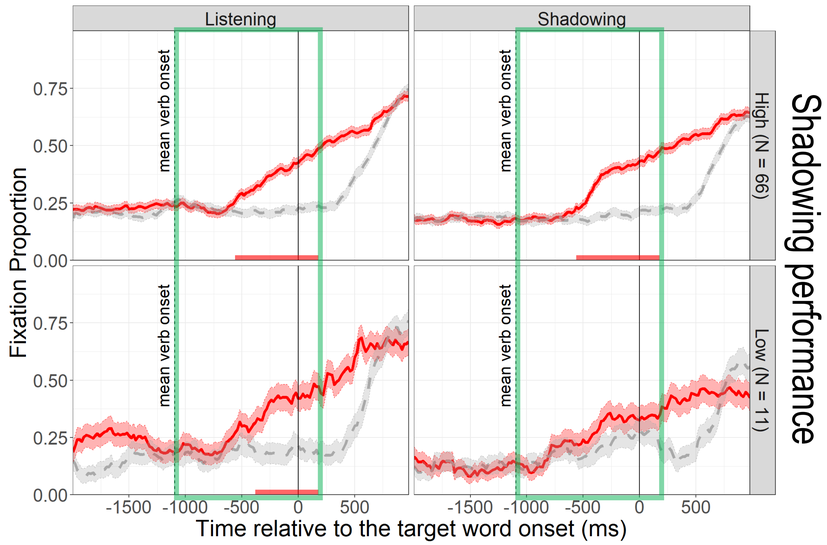
Listening: Listen to the sentence

Shadowing: Listen + Repeat the sentence as quickly and accurately as possible

Results



- Robust predictability effects in both tasks
- No predictability × task interaction
- Most participants shadowed sentences well
- Poor shadowers showed no predictability effect in the shadowing task



Conclusions

- No evidence that overt production facilitates prediction
- Shadowing interfered with prediction in poor shadowers ➔ Shadowing may have facilitatory and interfering effects, which may have cancelled each other out in good shadowers
- The verb-based prediction may not rely on the optional prediction-by-production mechanism

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